

TiLiA – TimeLineAnnotator

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Summary

‘TiLiA’ (TimeLine Annotator) is an open-source, cross-platform network of tools and resources designed to facilitate real-time creating, displaying and interacting with timeline-style annotations of audio, video and musical scores. Developed in Python with the PySide UI framework, TiLiA is available primarily as a desktop Graphical User Interface (GUI), supported by a Command Line Interface (CLI) as well as a free and open online platform where users may upload, view and query existing analyses.

Statement of need

Visual analyses of music have proven to be an enduringly popular, effective, and versatile tool for conveying a variety of musical ideas.¹ As the famous adage goes: ‘a picture is worth a thousand words’. A comprehensive analysis of music requires being able to summarise the relationships between events that are happening concurrently and through time. Furthermore, whatever potential a *static* image has, an *interactive* one offers considerably more. Form analysis, for instance, can be hard to parse (especially in real-time) but is often easy to understand through visual summaries.

In today’s digital age, we have come to expect certain basic interactive features across visual interfaces, such as resizing content to fit the window, keyboard shortcuts, undo/redo functionality, and so on. Musical contexts extend this with additional expectations, such as real-time synchronisation (a.k.a., audio or score following).

Despite several emergent initiatives (see below), musical scholarship has not yet made the leap into this technological space with any degree of decisiveness or consistency. Even most visually minded music analyses are consigned to static images embedded in print (or in e-copies at best) and not subject to the expectations for *fair*, open-source data sharing that are now *de rigueur* in other fields.

State of the field

While one *can* appropriate score and audio editors for visual analysis (Peebles, 2013), they are limited in their annotating capabilities. Dedicated tools fall into a few groups, according to their focus. Audio-analysis tools such as Sonic Visualiser (Cannam et al., 2010) and Partiels (Guillot, 2025) centre on audio and plug-in features, and lack score integration. Score-annotation platforms such as *Edirom Online* (Röwenstrunk, 2014), *Listen Here!* (Weigl et al., 2023) and *mei-friend* (Goebel & Weigl, 2024) anchor annotations to notated elements rather than to time and lack dedicated types for analytical structures such as form, while others, such as the MPM Toolbox (Berndt, 2021), are tied to specific file formats. “Timeline-based” annotators such as

¹See Isaacson (2023) for a survey of this long history.

39 *Audio Timeliner* (Notess & Swan, 2004; Yorgason, 2018), *iAnalyse* (Couprie, 2019), *BriFormer*
 40 (Jarvis, 2020) and *Dezrann* (Giraud et al., 2018) tend to focus more on providing tools for
 41 rich hand-made analysis.² While each group excels at particular tasks, there remains space for
 42 a solution that caters to more cross-cutting needs.

43 Platform support, for instance, is often narrow (Table 1): *Dezrann* and *BriFormer* run only
 44 in the browser, where local files are impractical and media face copyright restrictions, while
 45 *iAnalyse* runs only on macOS. Few of the tools are scriptable, and most store analyses in
 46 tool-specific formats, constraining interoperability and work at scale.

47 Most software focuses on a subset of the many guises analytical annotations can take
 48 (e.g. hierarchical groupings, metre, overlapping time spans, score elements, harmony, text).
 49 *Audio Timeliner* and *BriFormer* allow only nested spans, whereas *Sonic Visualiser* and *iAnalyse*
 50 allow overlaps, but only on a flat surface. Datasets need all of them: *SALAMI* (Smith et
 51 al., 2011) and the *Wagner Ring Dataset* (Weiß et al., 2023) encode hierarchical structure,
 52 itself an active research topic (McFee & Ellis, 2014); the *ABC* (Neuwirth et al., 2018) and
 53 the *Annotated Mozart Sonatas* (Hentschel et al., 2021) layer harmony, phrase and cadence
 54 labels; *BPSD* (Zeitler et al., 2024) and the *Harmonix Set* (Nieto et al., 2019) align formal
 55 structure with beats and measures; and the *Schubert Winterreise Dataset* (Weiß et al., 2021)
 56 links harmony annotations to scores and several recordings.

57 Input formats and multimodality have varied support. Symbolic-focused platforms, for instance,
 58 rarely support video. The *Walküre* (Balke et al., 2017), *Freischütz Digital* (Müller et al., 2013),
 59 *Erkomaishvili* (Rosenzweig et al., 2020) and *Lohengrin TimeMachine* (Lewis et al., 2021)
 60 interfaces show what multimodal, synchronised and richly interactive access can offer, but are
 61 tied to specific corpora.

62 In response to this challenge, we present ‘TiLiA’: a timeline annotator for all. It provides an
 63 open-source, cross-platform desktop tool that aligns annotations with audio, video, YouTube,
 64 PDFs and scores on shared timelines, offers dedicated types for musical structure, harmony
 65 and metre, and can be scripted and exchanged through CSV and JSON.

Table 1: Comparison of music annotation tools.

Tool	Annotation types	Media	Open source	Scripting & interop	Platforms
TiLiA	H I O B C (E)	A V Y S P	GPLv3	CLI; CSV, JSON	W L M (web)
Audio Timeliner	H I T	A	—	CLI	W M
BriFormer	H I (O)	A Y	—	—	web
Dezrann	(H) I O C E	A Y S P	GPLv3	JSON	web
iAnalyse	I O E T	A V S P	—	—	M
mei-friend	I C E T	S P	AGPLv3	MEI	web
Sonic Visualiser	I O B C T	A	GPLv2	CSV, CLI	W L M

66 *Annotations*: **H** hierarchical · **I** instant · **O** overlapping · **B** beat · **C** harmony · **E** score
 67 elements · **T** text

68 *Media*: **A** audio · **V** video · **Y** YouTube · **S** score · **P** PDF

69 *Platforms*: **W** Windows · **L** Linux · **M** macOS

70 Parentheses mark partial support.

²Commercial platforms such as *Hookpad* and *Soundslice*, which are closed-source and subscription-based, are omitted, as well as *SyncPlayer* (Fremerey et al., 2007; Kurth et al., 2005), which is no longer available.

Specifications

TiLiA consists of an organisation of digital tools, primarily of its [open-source desktop application](#) and [command-line interface](#), but also of a supporting online platform. The desktop application is developed in Python using the PySide binding for the Qt UI framework. Automated testing is done with pytest, and fully-automated deployment is available via GitHub Actions. The code base is loosely organised around a Model-View-Controller (MVC) pattern, which allows both the CLI and the GUI to benefit from the same backend logic. We provide object-oriented base classes such as Timeline, TimelineUI and TimelineComponent as a means to promote extensibility and support an evergrowing range of annotation needs.

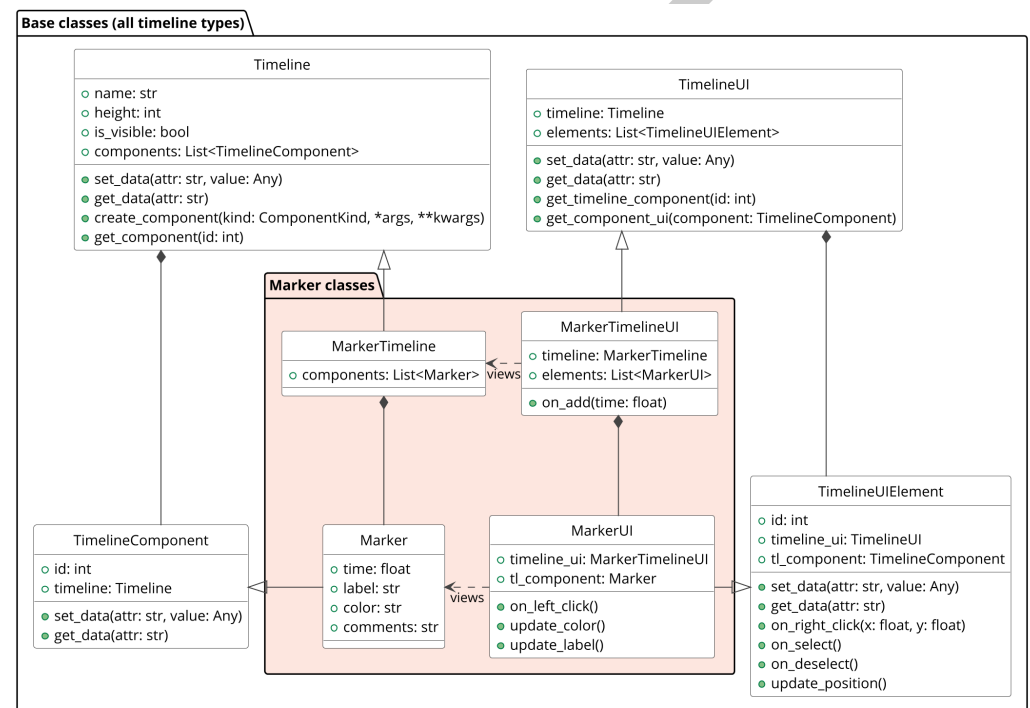


Figure 1: A simplified diagram for the main timeline base classes.

Figure 1 is a simplified diagram for the main timeline base classes. The outer box shows the base classes shared by all timeline types; the inner box shows the concrete subclasses for the marker timeline type, one example among the various timeline types. Timeline acts as container for TimelineComponents. Both act as models of TimelineUI and TimelineUIElement, which are views; in the marker example, MarkerTimelineUI and MarkerUI view MarkerTimeline and Marker respectively.

TiLiA is designed to facilitate the *creation* of analyses — particularly where this requires complex alignment with audio/video sources — and the (conversion and) *import* of analyses of external origin through CSV files. This effort thus connects TiLiA to wider datasets and scholarship, while also expanding its functionality. The [tilia-dcmli](#) repository demonstrates how the TiLiA CLI might be used to visualise data from an external corpus of musical analyses. The CLI can also *export* timelines created in TiLiA to a JSON file suitable for processing by other software.

93 **The desktop application**

94 The main part of TiLiA is a cross-platform *What-You-See-Is-What-You-Get* GUI (Figure 2),
95 designed to feel intuitive to anyone who has worked with audiovisual editing software. A
96 TiLiA analysis consists of timelines, each containing annotations that can be aligned to an
97 audio, video, YouTube stream or musical score. Each *timeline type* is designed to support
98 conventional formal annotations, with the affordance to work with a variety of other analytical
99 styles. These timelines each contain one or more *component types* (e.g., markers for marker
100 timelines), each with different *properties* to convey information including comments, colour and
101 labels.

102 We connect timelines and their components by using the loaded media as a “base layer” to
103 which all timelines refer. Timeline components have either *time* or *start_time* and *end_time*
104 timestamps which position them in relation to the underlying media. If a different media is
105 loaded, the user has the option to either uniformly scale timelines to the new media or to keep
106 them at their original position (deleting out-of-bounds components if necessary). If a beat
107 timeline is present, the software can map timestamps to metric positions, also providing a
108 metric-based interface for importing, exporting and interacting with timelines.

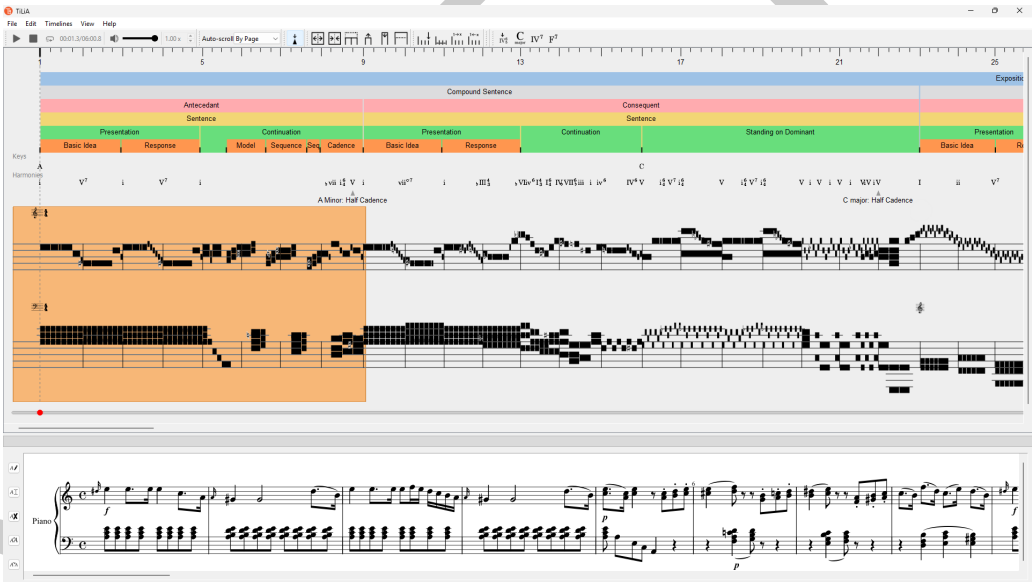


Figure 2: Excerpt of a TiLiA analysis on the desktop application.

109 Currently, there are eight types of timelines:

Timeline type	Purpose
Audiowave	Displaying amplitude graphs
Beat	Marking beats/measures, synchronising metrical and absolute time
Harmony	Chord symbols and roman numeral analysis
Hierarchy	Representing hierarchical relations and sequences
Marker	Annotating discrete timepoints
PDF	Synchronising external PDF documents with playback
Range	Representing non-hierarchical relations and sequences
Score	Synchronising digitised scores with playback

Command-line interface

In addition to the desktop GUI and the web app, a **command line interface** allows the automation of file and timeline creation and edition, media loading and external data importing. Sequences of commands can be loaded by the CLI from scripts written in plain text files, enabling programmatic conversion from various data formats to TiLiA files.

```
metadata set-media-length 120
timelines add hierarchy --name "Form"
timelines add marker --name "Formal closures"
timelines add hierarchy --name "Harmonic segments"
timelines add beat --name "Measures" --beat-pattern 1
timelines add marker --name "Harmony"
timelines add marker --name "Harm. functions"
timelines add marker --name "Keys"

load-media <path to audio> --scale-timelines yes
metadata set title "Title"
save <path to TiLiA file> --overwrite
export <path to JSON file> --overwrite
```

Online platform

The TiLiA desktop application is supported by an [online platform](#) that lets users store, share, visualise and even query existing analyses. Registration is free and is only required for uploading content. This reflects a goal of TiLiA to support not only the modernised creation and storage of analyses, but also to support the still-rare practice of analytical *collaboration* and structured *searches* on the corpora; both which benefit from network effects facilitated by the lowered entry barrier.

Outlook

The TiLiA desktop application aims to provide a user-friendly annotation tool both for MIR researchers and musicologists. It can be used to create large annotation datasets or detailed, interactive analyses of individual pieces. It can also be used in the classroom to better connect theoretical concepts with audiovisual material or as a common interface for analytical assignments.

In the future we hope to convert existing open-source datasets and integrate them into the web platform, providing a centralised hub where diversified audiovisual data can be queried and interacted with in a uniform manner. We also plan to improve audio processing and score displaying capabilities and to add more timeline types (e.g. a 2-dimensional graph timeline) to offer more diversified visualisations.

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